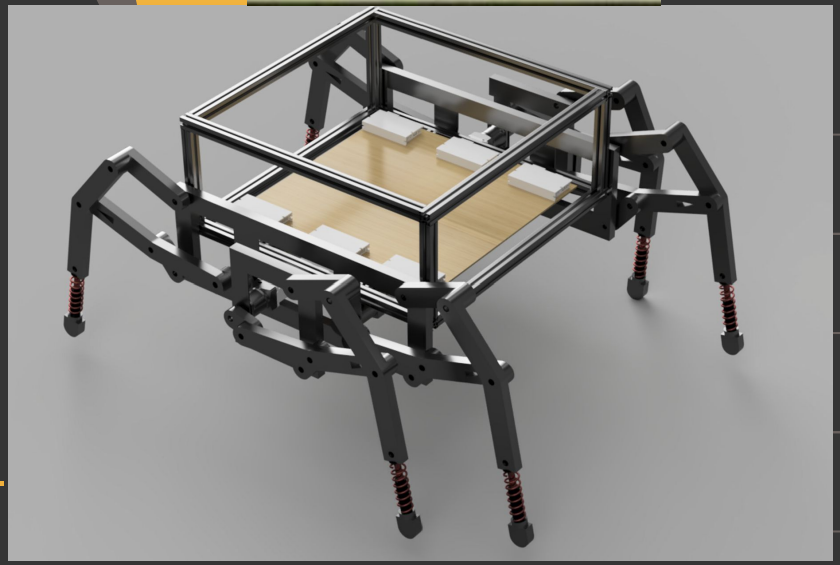
Dancing Hexapod
Team 2
Mid Semester

Yechen Zheng



Design requirements

$1000 budget for assembly
$2500 budget for motors, drivers, encoder, battery

Weigh less than 150 lb, while carrying more than 170lb
No discernible damage when dropping 25 lb loads on top

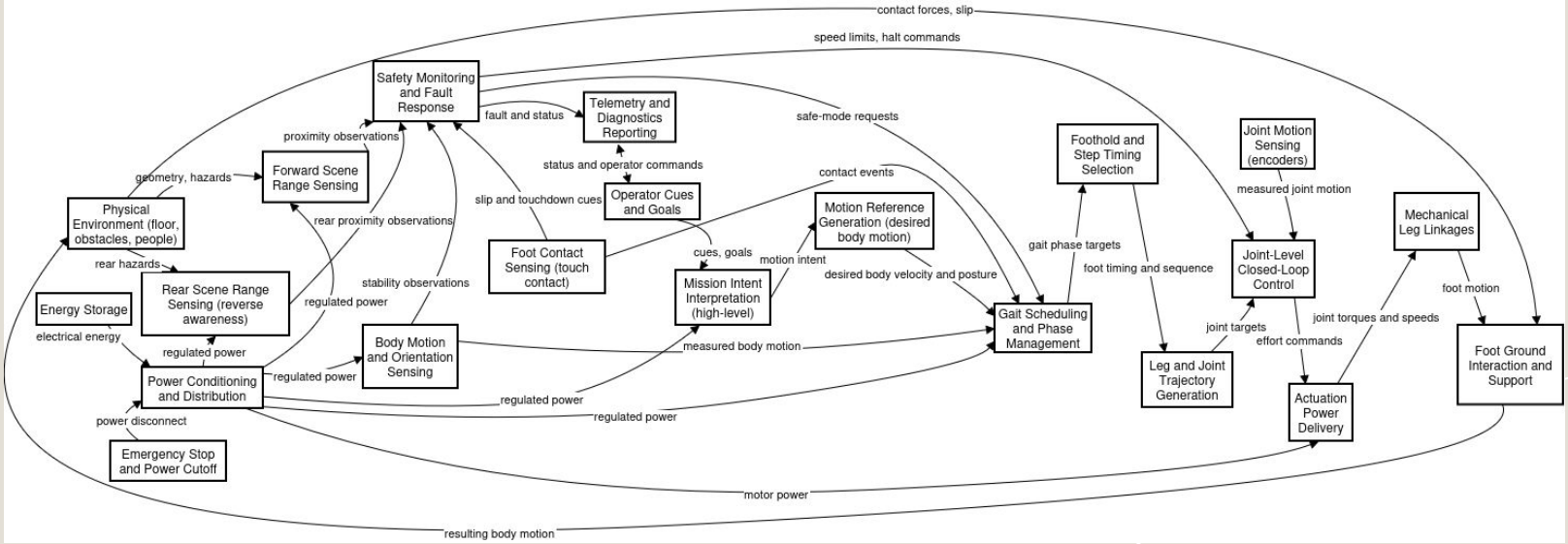
Legs must drive locomotion (aux. Wheels allowed support only, no power/odom)

Ability to move forward, backward, and turn either direction with zero radius

At least 30 min battery life in worst-case motion

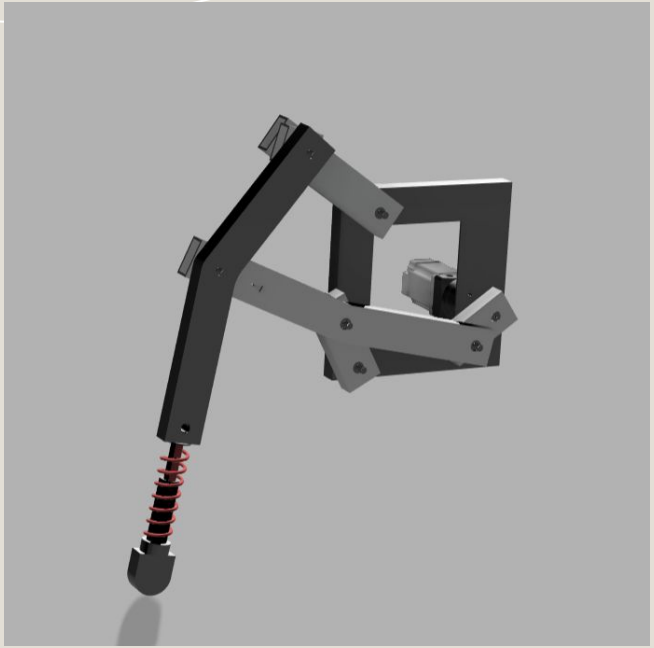
Maximum: 30 seconds for 10’ forward motion
45 seconds for 10’ forward curvilinear - unknown angle!
45 seconds for 10’ backward motion
60 seconds for 10’ backward curvilinear - unknown angle!

Functional architecture

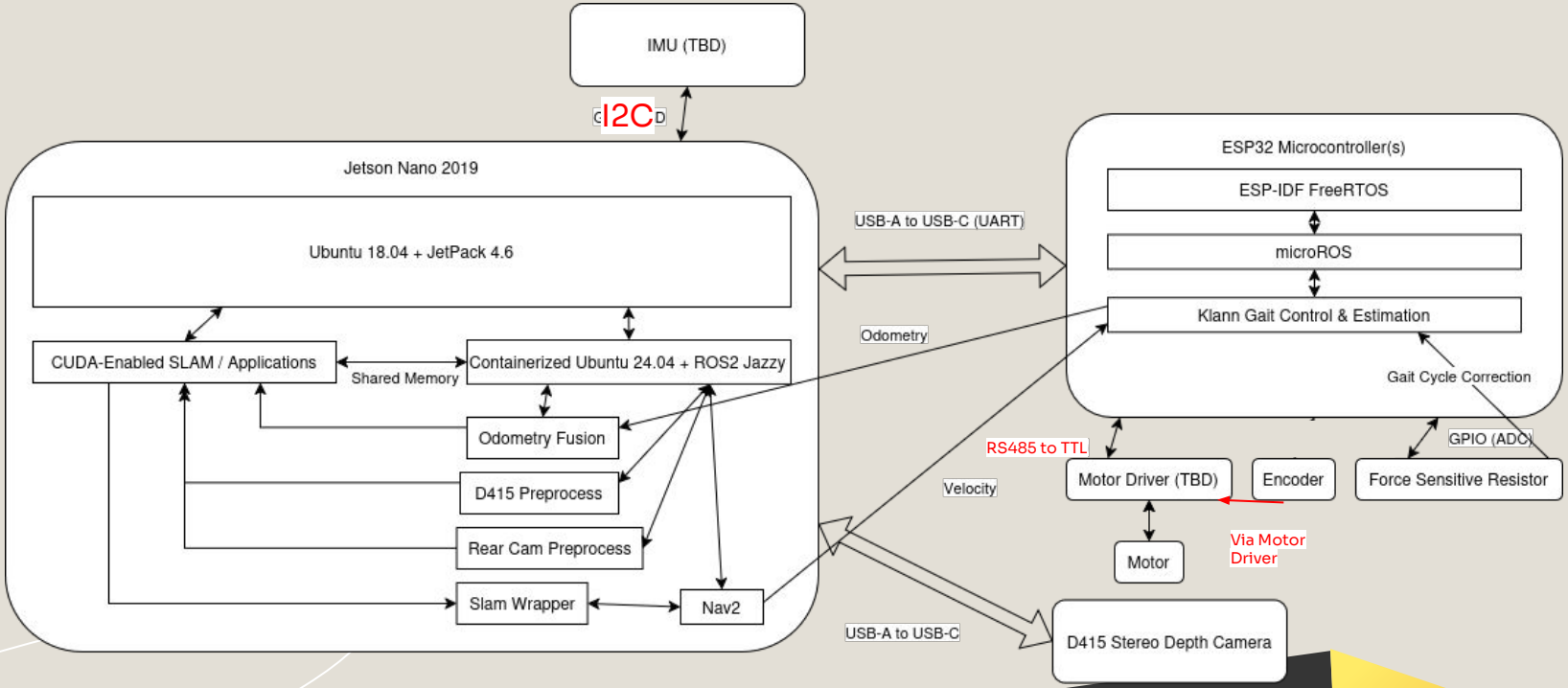


Design concept – Locomotion

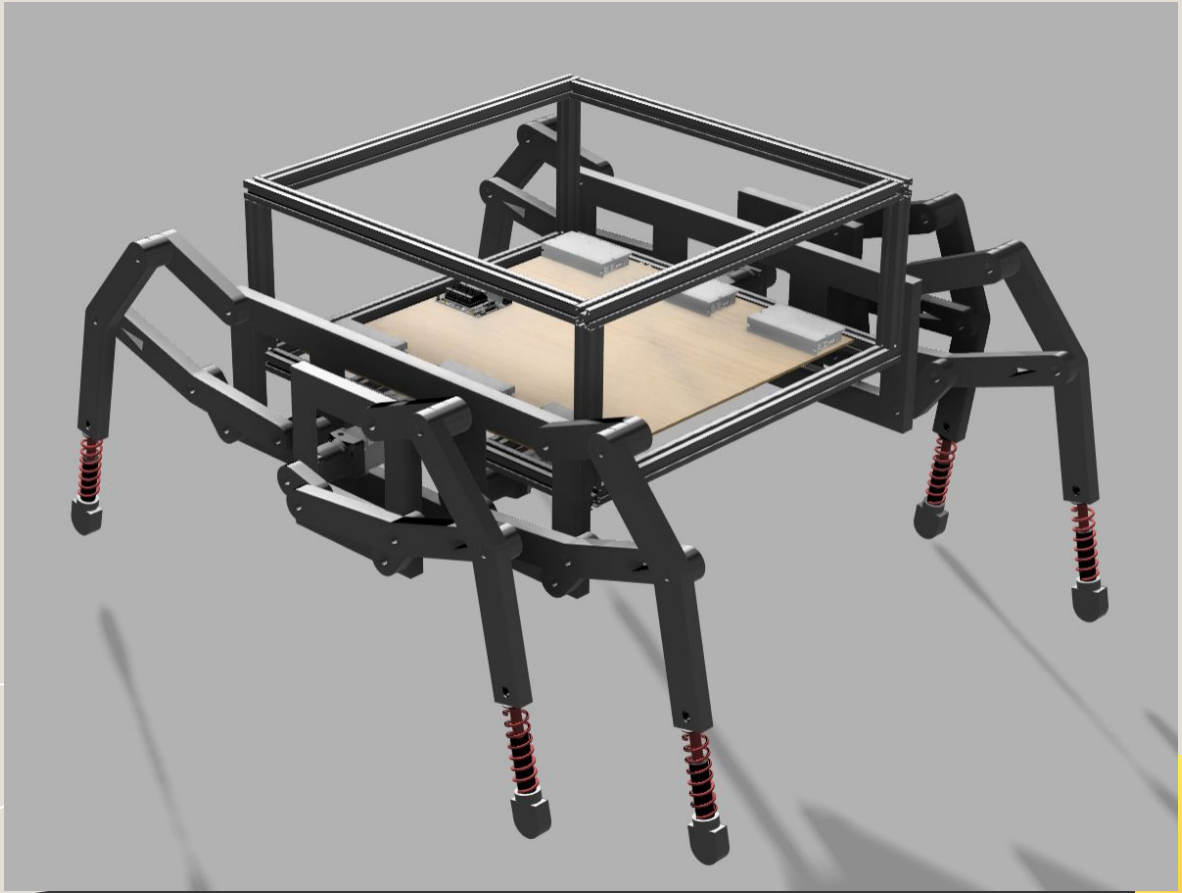
	Our Design	Others Design
Locomotion	Klann linkage	Fully actuated legs
Actuation	6 motors (updated)	Multiple motors per leg
Architecture	Mechanically coupled → now independent motors	Independently actuated
Gait generation	Mechanical → now hybrid (mechanical + control tuning)	Software-defined tripod gait
Compliance	Passive springs	None



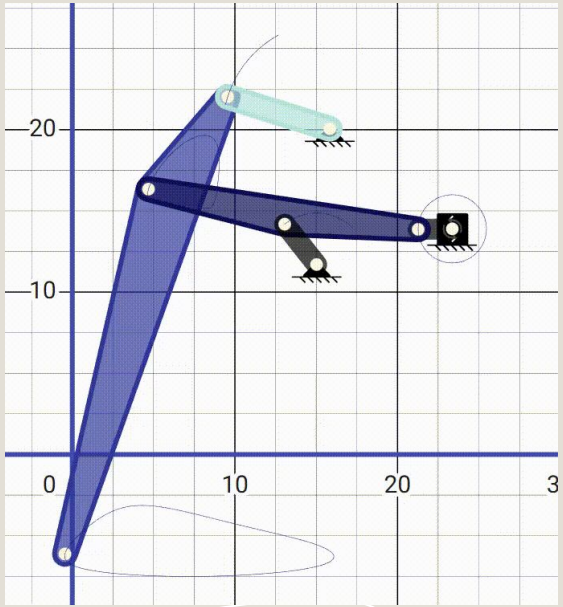
Cyberphysical architecture

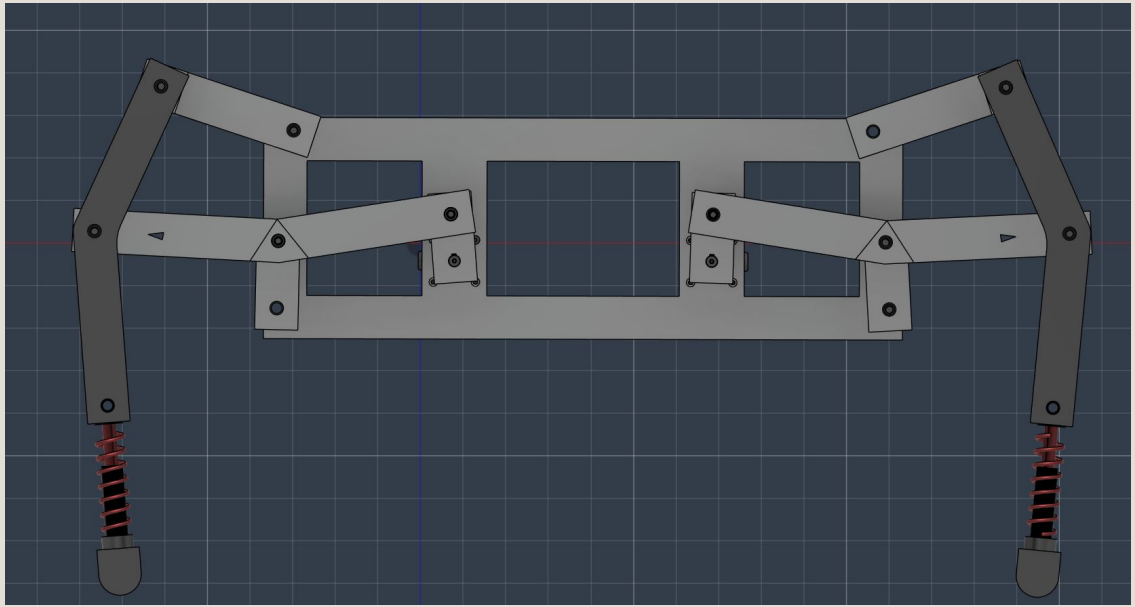


System depiction - Overview

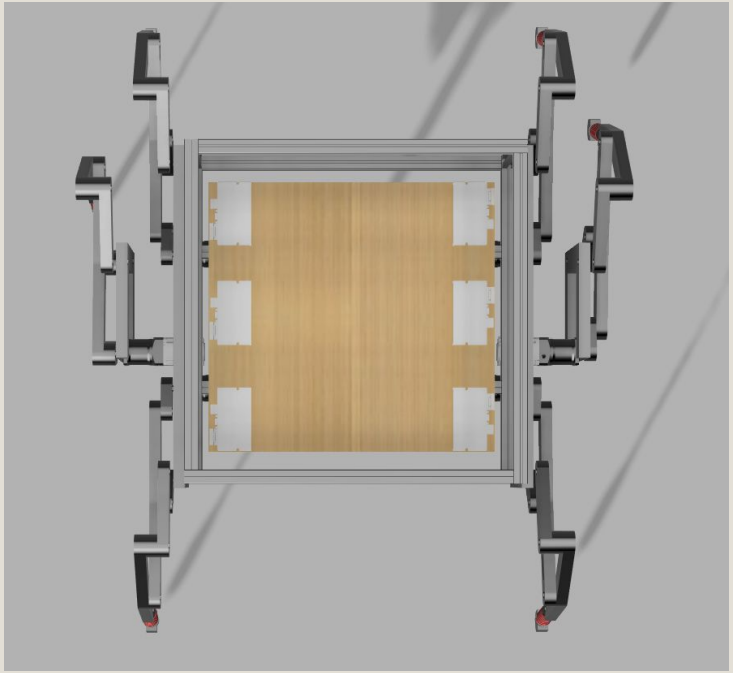


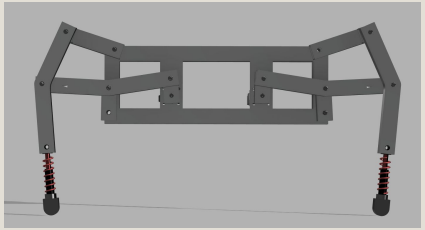
System depiction - Klann Linkage

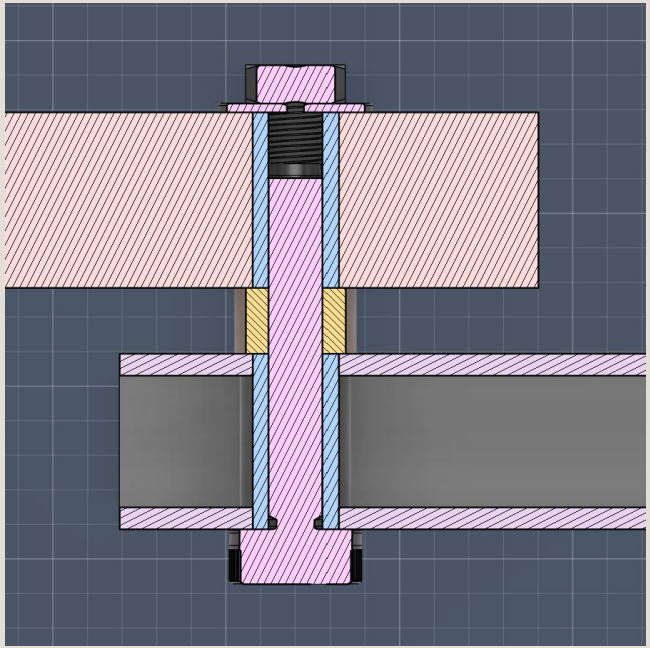




CURRENT STATUS - ASSEMBLY - LEG



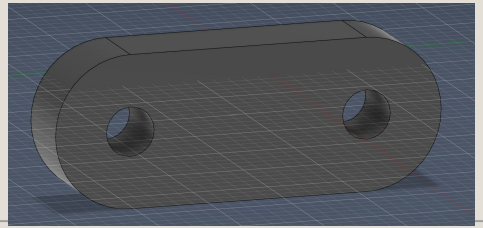
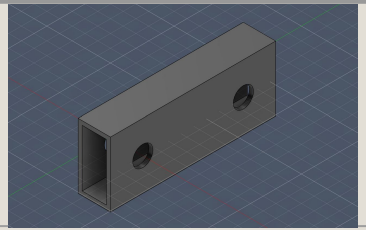
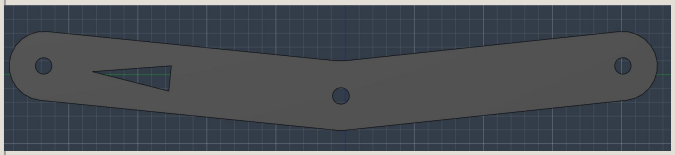
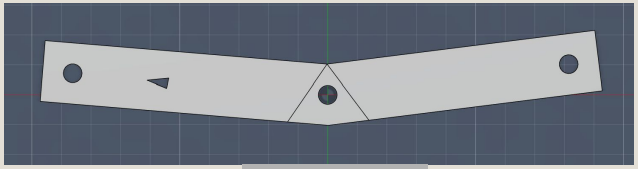
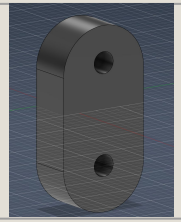
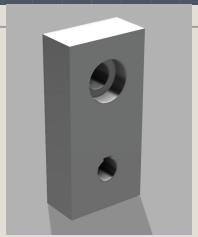




- Updated leg assembly with redesigned parts
- Specified linkage connection method

CURRENT STATUS - MANUFACTURING

- Parts have been redesigned for improved manufacturability, utilizing standard stock materials such as rectangular aluminum tubing

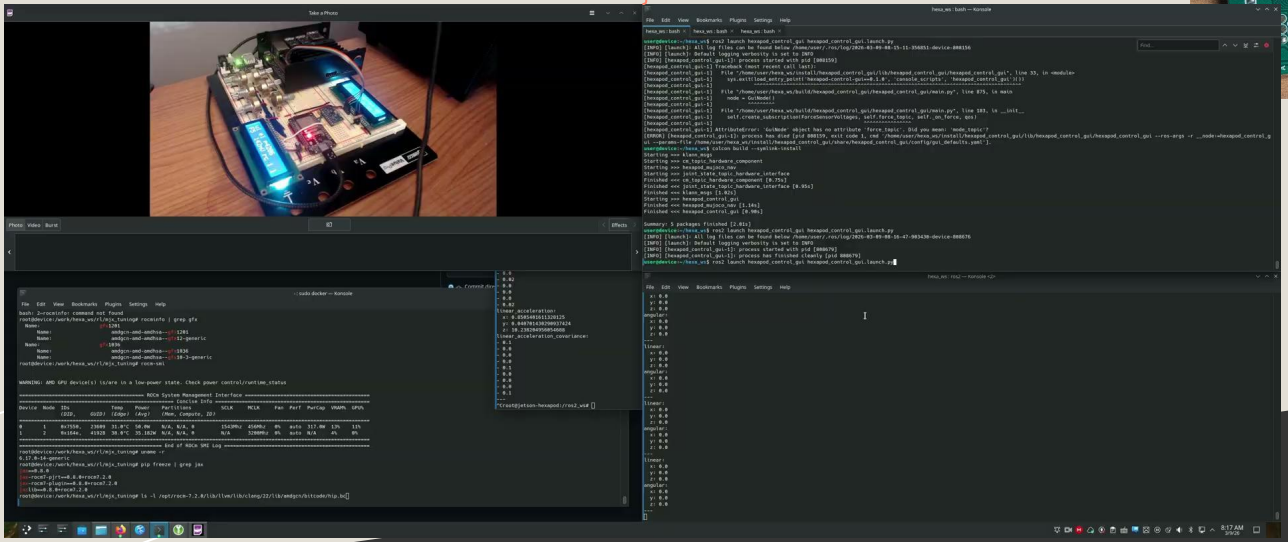
Before	After
	
	
	

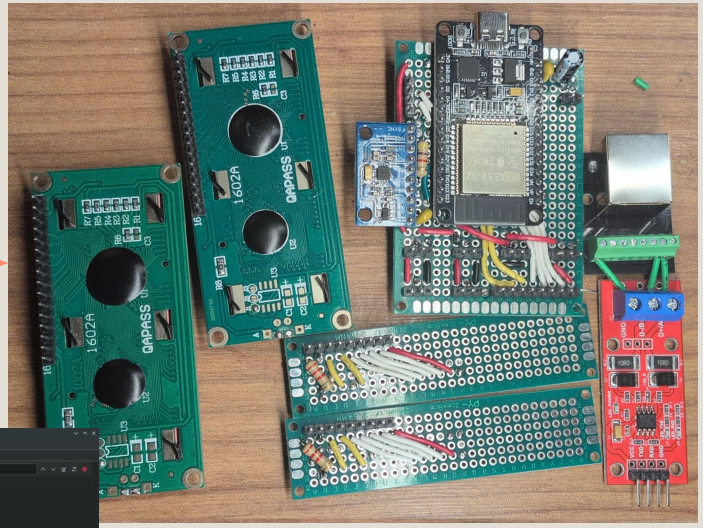
CURRENT STATUS - Motor and Gearbox

- Rated Voltage: 48VDC
- Rated Torque: 1.27Nm
- Peak Torque: 3.81Nm
- Rated Current: 10.2A
- Rated Speed: 3000rpm

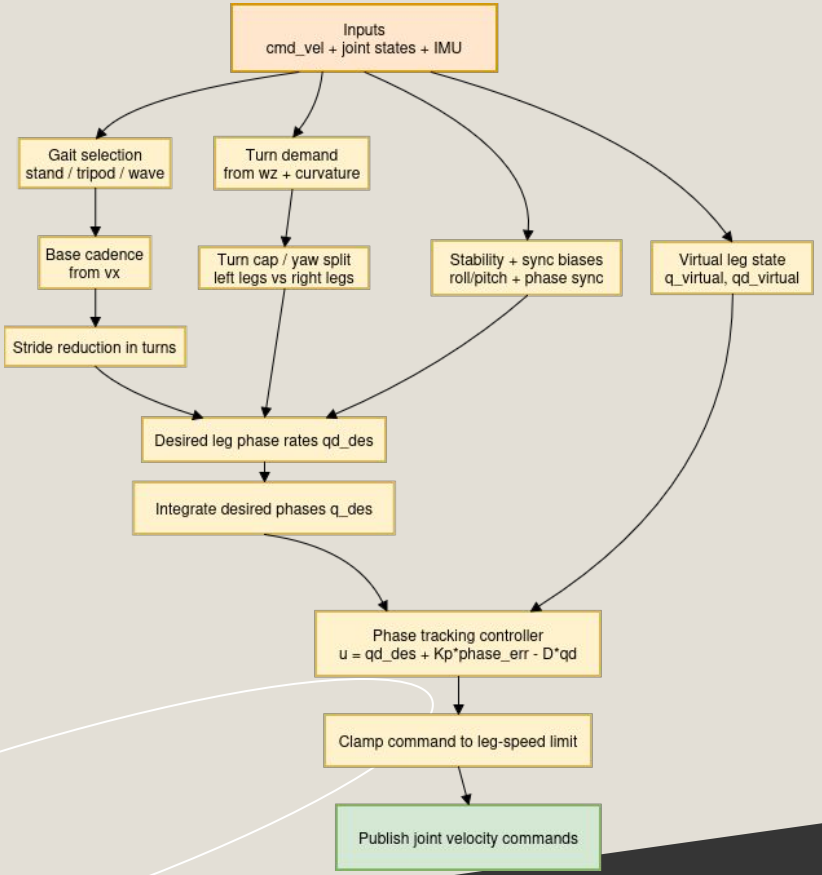
- Gearbox Type: Spur Planetary
- Gear Material: Alloy Steel
- Gear Ratio: 50:1

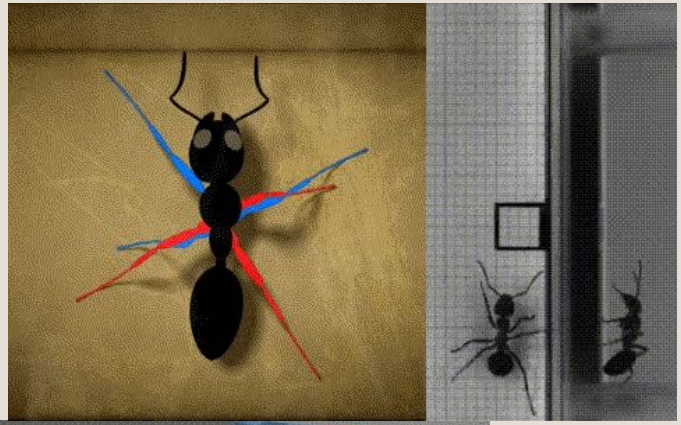
CURRENT STATUS - CIRCUIT

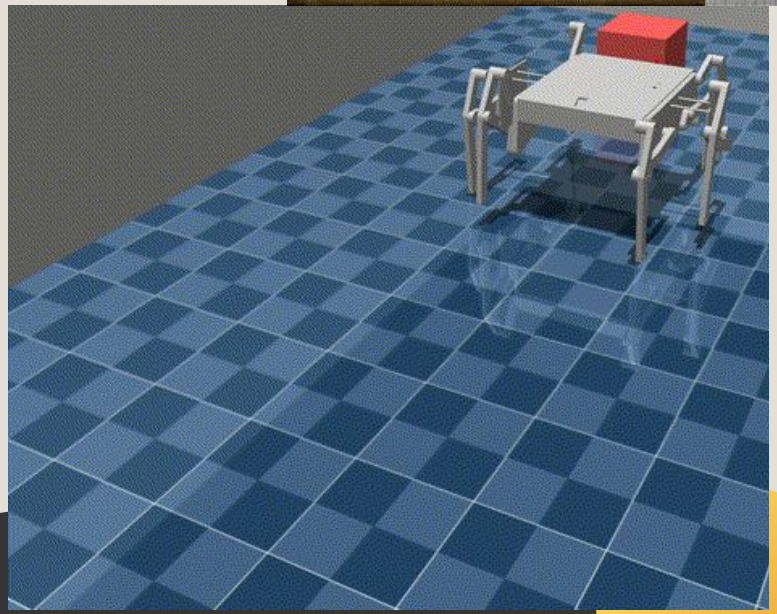




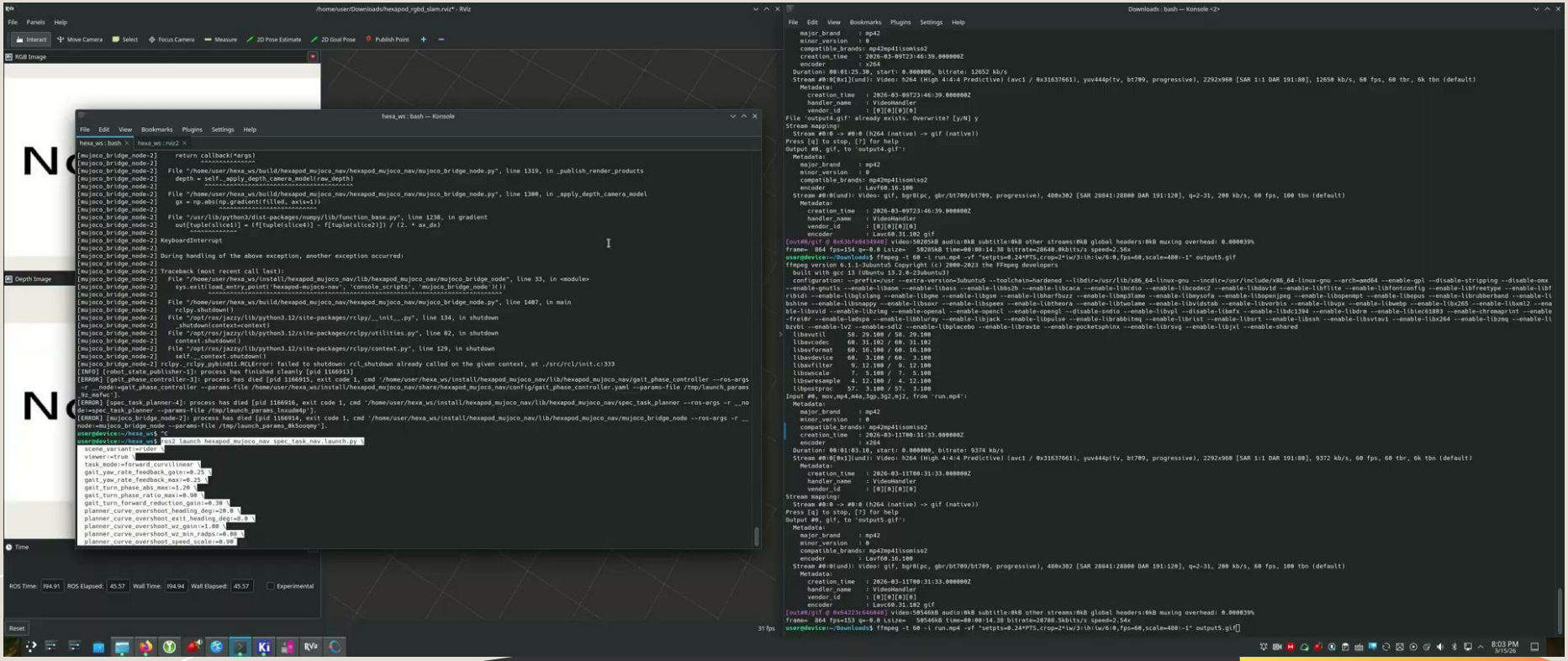
CURRENT STATUS - CONTROL







CURRENT STATUS - CONTROL



Schedule & Responsibilities

System Demo 5 1				
15	SD5 Demo Plan + Artifacts #71		Todo	1 / 10 10%
	Finish chassis cage and mount sensing / compute without power system #80	dgl7, hongfeil-bee...	In Progress	
	Port over optimized parameters to CPG in-ros controller #81	dgl7	Done	
	Finalize power circuit and place part orders #13	dgl7	In Progress	
	Finalize RS485 motor driver in ESP32 and run thread #82	dgl7	In Progress	
	Test motor driver in oscilloscope pending motor arrival #83	dgl7	Todo	
	Finish setting up laptop ubuntu / ros2 dual boot to be able to test deployed sys... #85	dgl7 and yechenz-wq	In Progress	
	Construct single leg according to most recent design #87	hongfeil-beep and li...	Todo	
	Add all nav2 components to simulation for set-point navegation, mode selectio... #84	dgl7	In Progress	
	Add touch sensing to legs of robot in simulation to polish controller phase #86	dgl7 and yechenz-wq	Todo	
	deploy slam package to jetson and test with rviz + laptop wifi #88	dgl7 and yechenz-wq	Todo	
+ Add item				
System Demo 6 1				
16	SD6 Demo Plan + Artifacts #72		Todo	0 / 5 0%
	Mount and drive single leg using esp32 controller #89	dgl7, hongfeil-bee...	Todo	
	Implement optimized local planner that is jetson-runnable in simulation #90	hongfeil-beep, liu3...	Todo	
	Order remainder of leg parts #91	hongfeil-beep, liu3...	Todo	
	Test several SLAM packages and in-robot IMU odometry #92	dgl7	Todo	
	Install and verify power system #94	dgl7	Todo	
+ Mount and drive single leg using esp32 controller				
System Demo 7 1				
17	SD7 Demo Plan + Artifacts #73		Todo	0 / 4 0%
	Drive 6 legs successfully #93	dgl7, hongfeil-bee...	Todo	
	Port full ros2 control stack into microcontroller #95	dgl7	Todo	
	Port local planner into jetson #96	dgl7	Todo	
	verify gait controller / planner functionality, test e-stop features #97	dgl7	Todo	

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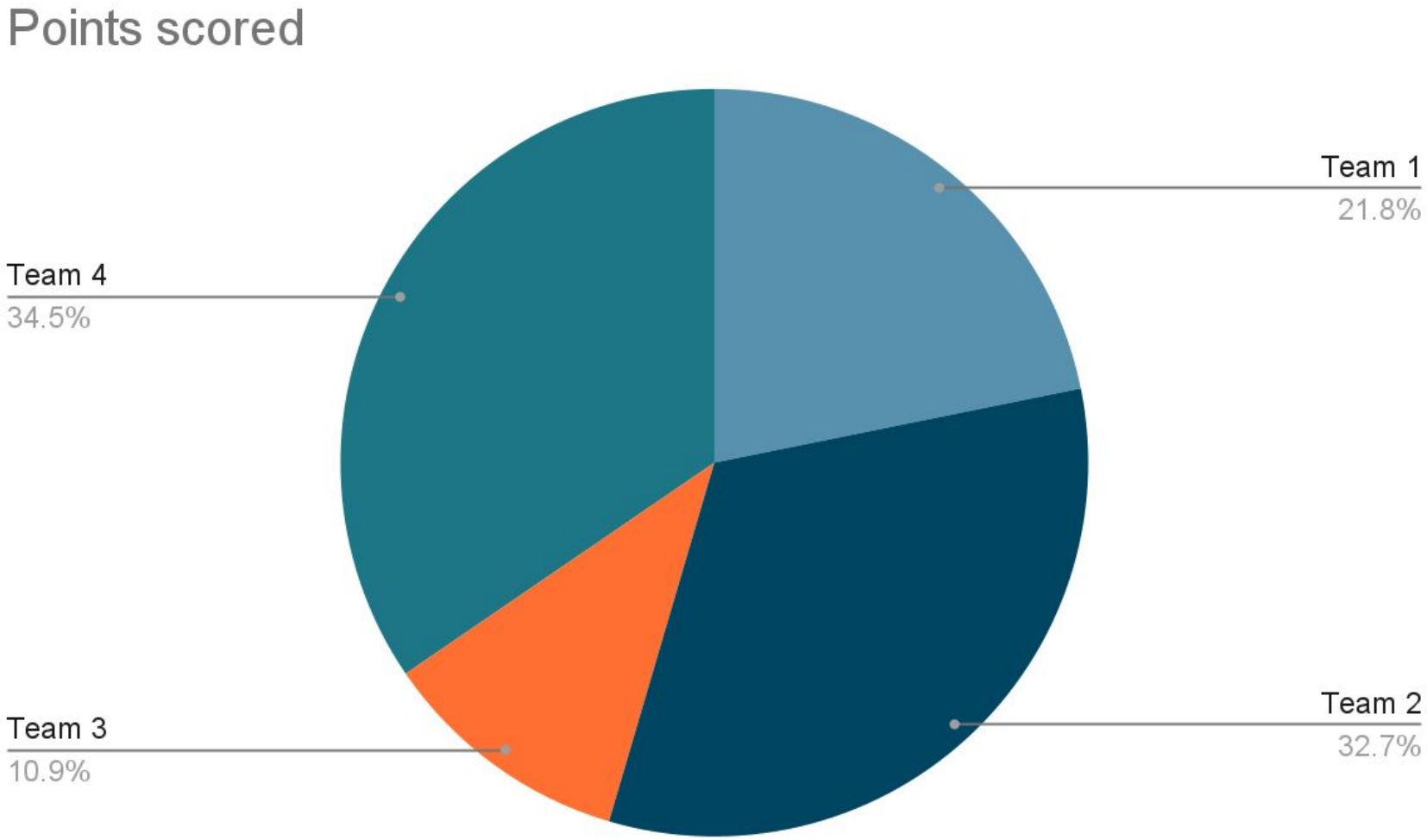

Current/Past Orders - Mechatronics Budget

Item Name	Vendor	Qty	Price	Status
Single Four Slot Rails 1" x 1" (700mm)	80/20	7	$101.99	received
Single Four Slot Rails 1" x 1" (670mm)	80/20	2	$28.16	received
10, 20, & 25 Series 2 Hole - 18mm Slotted Inside Corner Bracket with Dual Support	80/20	10	$12.70	received
10 Series M6 Standard Slide-in T-Nut	80/20	10	$16.10	received
M6 Slide-in Economy T-Nut Block	80/20	10	$5.30	received
20Sets 10 Extrusion Brackets 1" x 1" 1010 Series Aluminum Extrusion Profiles Connector Set, 20pcs 10 Extruded Aluminum Corner with T Slide Nuts Bolts for 10 Series Extrusion Profiles Rail	Amazon	1	$20.88	ordered
Single Four Slot Rails 1" x 1" (678mm)	80/20	2	$28.42	ordered
1" X 1" X 3/16" HOT ROLLED BAR SIZE ANGLE ASTM A36（12inch）	Fastmetals	1	$6.82	ordered
1" X 1" X 1/4" ANGLE 6061-T6 ASTM B308 (12inch)	Fastmetals	1	$6.33	ordered
2" X 1" X .125 WALL RECTANGULAR TUBE 6061-T6	Fastmetals	2" x 5	$82.20	ordered
Rear Shock Absorber	amazon	1	$13.89	pending
		Total	$322.79	

Estimated budget - Mechanical

Item	Description / Notes	Qty	Unit Cost (USD)
Acquired Items	/	/	$322.79
5 more sets of legs	Links & Bearing & Connectors & Bolts	1	$650
Wood panels	Mount on frame to support people and electronics	2-3	$30-50
Motor Mount	Mount for motors	6	$40
Shock Absorber	Leg shock absorber	6	$84
Total	/	/	$1112.79 - $1132.79

Risk management plan

- **Side Leg Design Risk:** Although geometry is finalized, welding-induced length/bending errors may cause interference or instability → redesign to eliminate welding dependency

- **Manufacturing Challenges:** Seek support from TechSpark faculty to ensure fabrication feasibility

- **Budget Overrun Risk:** Potential cost increase → covered by team members & find spare materials in Techspark

- **Motor Mount Misfit:** Current design does not fit → resolve using rapid 3D printing iteration